Kobe Bryant Shot Selection

Predictive Modeling and Sports Analytics

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2026

Context & Project Objective

The Data-Driven Era in Sports:

- Analyzing a multidimensional dataset of shots taken by **Kobe Bryant** during his 20-year career to optimize strategy via *sports analytics*.

Project Objective (Binary Classification):

- **Target Variable:** `shot_made_flag` ($Y \in \{0, 1\}$)
- **1:** Successful shot (Made) | **0:** Missed shot (Failure)
- **Goal:** Train Machine Learning algorithms to calculate the probability of a successful shot based on an input feature vector.

Dataset Description

Variables grouped into three main categories:

- **Spatial and Geometric:** Cartesian coordinates (X, Y) of the release point and distance from the basket.
- **Temporal and Contextual:** Time remaining, game period, season, opponent and home court advantage.
- **Biomechanical and Technical:** Specific offensive action performed (e.g., *Jump Shot, Layup*) and shot combination type.

Data Isolation & Target Distribution

Strict Data Isolation:

- Visual analysis strictly limited to the **Training Set** to prevent *Data Leakage*. Validation and Test sets act as secure "vaults".

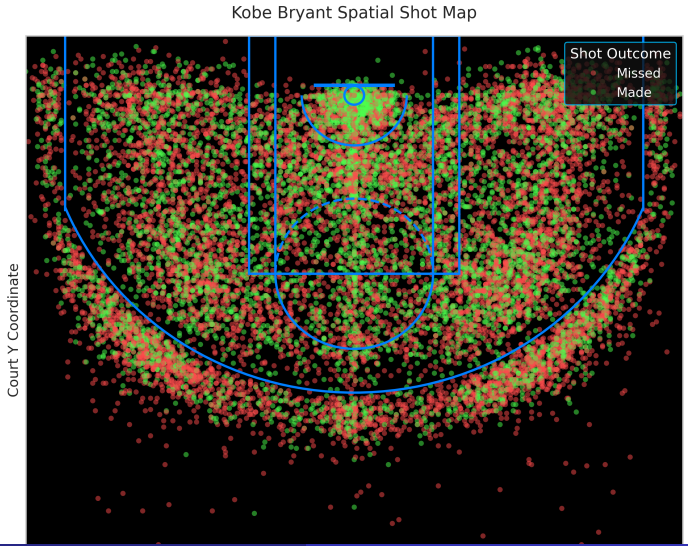
Class Imbalance Check:

- The dataset is **sufficiently balanced** (slight natural prevalence of misses).
- Eliminates risk of predictive bias; standard metrics (Accuracy) remain reliable.

Spatial Analysis: Kobe Bryant Shot Map

Key Insights:

- Mapped using continuous Cartesian coordinates.
- **Clusters:** High tendency to shoot from the paint and "elbows".
- **Efficiency:** High success near the rim; variance increases with distance.



Preprocessing & Feature Engineering

Missing Values Check: Zero missing data across all splits (`isnull().sum()`).

Advanced Feature Engineering:

- **Temporal:** `is_clutch_time`, `is_buzzer_beater`.
- **Contextual & Spatial:** `is_home`, `shot_angle` (visual perspective).
- **Matrix Cleanup:** Removed redundant (`lat`, `lon`) and ID variables.

Categorical Encoding: Target Encoding

Why Target Encoding?

- Avoids massive sparse matrices (One-Hot Encoding).
- Replaces textual categories with historical probability of success.

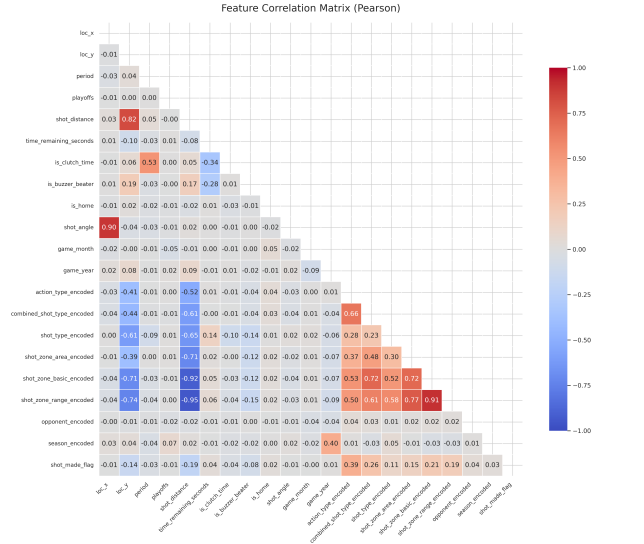
Strict Implementation:

- Calculated **exclusively** on the Training Set.
- **Fallback Mechanism:** Unseen categories assigned the `global_mean` of the training set to prevent errors.

Correlation Analysis

Pearson Matrix:

- Checked linear relationships between engineered features.
- **Result:** Some features deleted.
- Ensures stability of classification algorithms.



Tree-Based Models: RF & XGBoost

Random Forest (Baseline - Bagging):

- Minimizes *Gini Impurity*: $G = 1 - \sum p_i^2$
- Robust against overfitting, excellent interpretability.

XGBoost (Boosting):

- Builds sequential trees correcting residual errors.
- Optimizes Log-Loss + Regularization (Ω): $\mathcal{L}^{(t)} = \sum l(y_i, \hat{y}_i) + \Omega(f_t)$

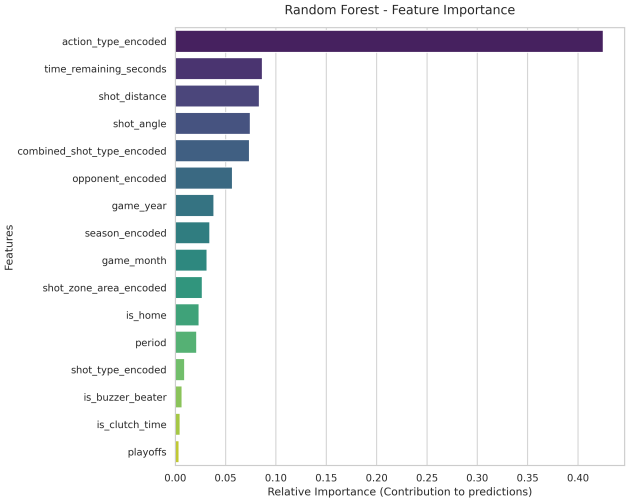
Hyperparameter Optimization:

- `RandomizedSearchCV` with 3-fold CV. Metric optimized: **ROC AUC**.

Feature Importance Analysis

Key Insights (Random Forest):

- Biomechanical execution (`action_type`) dictates outcome more than spatial position (∼**50%** combined weight).
- Temporal pressure (`time_remaining`) ranks 2nd, ahead of raw distance.
- Validates the extensive feature engineering phase.



Model Selection & Metrics

Validation Baseline Comparison:

- **Random Forest:** Accuracy 0.6791 | ROC AUC 0.6920
- **XGBoost:** Accuracy 0.6750 | ROC AUC 0.6947
- **Champion Model: Random Forest** (Chosen for interpretability).

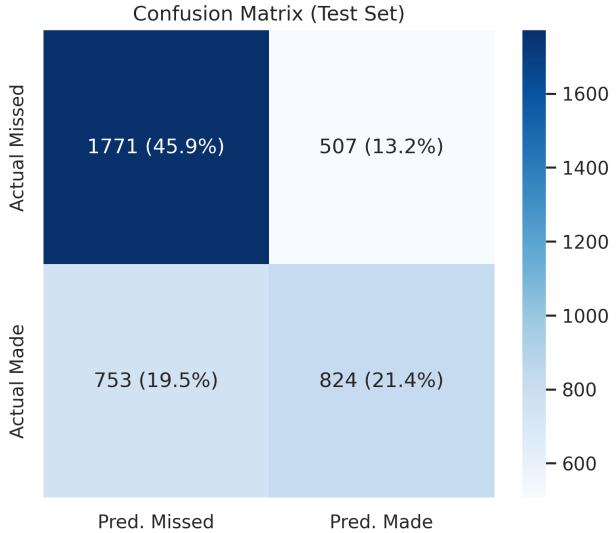
Final Test Performance ($N = 3855$):

- **Accuracy:** 67.32% | **ROC AUC:** 0.6956
- Metrics align with validation results, confirming **no overfitting**.

Final Test: Confusion Matrix

Recall Asymmetry:

- **Missed (0):** High Recall (78%). Model confident in misses.
- **Made (1):** Conservative behavior. Low Recall (52%).



Qualitative Error Diagnostics

Spatial Analysis of Errors:

- Avg distance (Incorrect): 14.87 ft. | Avg distance (Correct): 15.17 ft.

The Mid-Range Problem:

- Algorithm struggles as distance increases into the **mid-range**.
- **Why?** Tactical uncertainty heavily influenced by defender pressure.
- Contextual defensive data is absent from dataset, creating a natural ceiling for predictive reliability.

Project Summary & Limitations

Objective Achieved:

- Robust ML pipeline (Target Enc., isolation) validating domain-specific feature engineering.
- Champion Model (RF) achieved **67.32% Accuracy** on unseen data.

Domain Limitations:

- Drop in recall for successful shots and higher errors in mid-range.
- **Root Cause:** Absence of advanced situational metrics (the surrounding defensive environment).

Future Work

To overcome limitations and improve accuracy:

- **Integration of Tracking Data:** Incorporate optical tracking (defender distance) to quantify defensive pressure.
- **Sequential Modeling:** Implement RNNs to capture rhythm, fatigue, and the "hot-hand fallacy".
- **Deep Learning:** Train DNNs on expanded feature sets for complex non-linear biometric interactions.